

Using Speech Analysis in Voice Communication

A new approach to improve Air Traffic Management Security

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Abstract—This document describes the concept of an air traffic management security system and current validation activities. This system uses on one hand speech analysis techniques to verify the speaker authorization and measure the stress level within the air-ground voice communication between pilots and air traffic controllers; on the other hand it monitors the current air traffic situation. The purpose of this system is to close an existing security gap by using this multi-modal approach. First validation results are discussed at the end of this paper.

Keywords—air traffic control; voice communication; ATM security; GAMMA Project; speaker verification, stress detection

I. INTRODUCTION

In the last decades, the air traffic has been growing significantly. Especially the airspace of North America, Southeast Asia and Europe is highly frequented; major airports are close to its capacity limit. Air Traffic Control (ATC) services play a very important role to prevent collisions, to maximize airspace and infrastructure resources and to avoid delays.

Air Traffic Control services can be subdivided into [1]:

- Area control service: provision of ATC service to controlled flights enroute,
- Approach control service: provision of ATC service to controlled flights during arrival and departure phase,
- Aerodrome control service (Tower): provision of ATC service to controlled flights during take-off and landing.

Air traffic controllers (ATCos) worldwide work according to standard procedures and use ATC clearances to guide the air traffic in a safe, orderly and expeditious manner [2]. These clearances are still mainly communicated via analogue radio telephony, which is neither encrypted nor otherwise secured and which can easily be listened in or even intruded. Although the participation to ATC voice communication is only allowed for authorized persons, there is no instrument in place which verifies this authorization. Some incidents in the past showed that it is easily possible to insert fake ATC clearances by any intruder, which could also be used to provoke conflicts and collision risks between two airplanes or at least to disturb the safe, orderly and expeditious flow of air traffic significantly, causing an economic loss [3].

This paper describes the conceptual solution that is developed in the frame of the Global ATM Security Management Project (GAMMA Project) by the Slovak Academy of Science (SAV) together with the German Aerospace Center (DLR). The concept uses a multi-modal setup of detection functions and its features can be seen as an artificial cognitive capability as defined in [4]. A short summary of planned and ongoing validation activities is also included.

II. APPROACH – COMBINATION OF MODALITIES

To close this security gap in ATC voice communication, a system is developed which combines several vocal and non-vocal monitoring functions. This system is called “Secure ATC communication” (SACom) and is designed as a pure detection system, because other functionalities such as suppression of malicious content, encryption or adding security signatures would make a direct intervention in the existing ATC voice communication necessary, which cannot be brought in line with existing safety regulations and which is also prohibited by the basic prototype requirements of the GAMMA project [5].

The SACom system uses four independent modalities to assess the present security status of ATC voice communication. Each modality is a separate analysis process which produces an individual indicator. These indicators are correlated finally. The basic assumption is that several independent indicators together enable a more reliable and more flexible security assessment covering a broader range of possible attack scenarios.

The four modalities that are used by the SACom system are:

- Speaker verification (voice communication analysis),
- Stress detection (voice communication analysis),
- Conformance monitoring (non-vocal analysis),
- Conflict detection (non-vocal analysis).

These modalities are explained in more detail in the following chapters.

The SACom system is designed for use in controller working positions (CWPs) and in aircraft cockpits. It needs access to air traffic surveillance data and given ATC clearances, which are assumed to be available in the present or

in the near future since the use of datalink communication and system-wide information management (SWIM) is rapidly increasing [6]. Beside direct display of corresponding indicators to the user (the air traffic controller or the pilot), the SACom system is able to automatically send an alert to any connected security operation center. The GAMMA project also develops a national/international instance of such a security operation center, called Security Management Platform (SMP).

At present time, a prototype for the SACom system is already available and about to be validated.

III. VOICE COMMUNICATION ANALYSIS

The automatic analysis of human voice communication and speaker verification, has already found its applications in voice operated services and in biometrics. The last decade brought even more robust algorithms and enough computing power, which enabled a wide use of speaker verification in commercial security systems.

Another promising area of research is the stress and emotion detection from voice.

The authors believe that a rapid and reliable speaker authorization in the voice communication channel and early detection of stress in pilots and air-traffic controllers could contribute to the ATM security.

A. Speaker Authorization

The "Speaker Authorization" module will be responsible for on-line monitoring and checking whether the speakers taking part in the "radio" voice communication belong to the actual list of the authorized speakers.

The use-case of speaker authorization considered in the GAMMA project is therefore rather different from that of standard verification due to the open-set character of the recognition task. It can be considered as a "Multi-target detection" [7] or "Open-set, text-independent speaker identification (OSTI-SI)" [8]. The given test utterance is assigned to the speaker model that yields the maximum similarity over all speaker models in the system, if this maximum likelihood score itself is greater than the threshold. Otherwise, it is declared as originated from a non-registered speaker [8]. This task with dozens of authorized speakers and unlimited number of potential intruders is really demanding. Very short utterances, noisy and narrow range communication channel, clipped or weak signal and non-native speakers make this task even more difficult.

The main objective in GAMMA is not to create a perfect speaker verification system, but rather to verify whether it is possible to apply this technology to enhance the ATM security. Therefore standard tools were used for development.

The speaker verification system was created using KALDI research toolkit [9]; the i-vector approach [10] was used with PLDA scoring [10]. LibriSpeech corpus [11] was used for UBM training (2500 speakers, 3 minutes of speech per each).

In the process of prototype validation there will be only a few speakers involved in the experiment and any real statistical evaluation will not be possible. Therefore a part of the VoxForge database [13] was used as a test set to statistically

check the classification properties of the speaker verification system. 400 speakers were enrolled to serve as "authorized" in authorization tests. The enrollment recordings consisted of several utterances giving approximately one minute of speech per speaker. The utterances were taken from different recording sessions when available. Other 1500 speakers were chosen as "impostors" (non-authorized). The total number of 80 000 test utterances was used with the length of 2 to 10 seconds.

In the first step the speaker verification function was checked. In the Speaker verification test each of the enrolled speakers has been tested against his own utterances (target) and against utterances of each of all the other speakers (non-target). The distance between a particular speaker model and the actual incoming utterance is expressed as score. The graph of score distribution on Fig. 1. shows that the verification works well on the VoxForge data, as the bells of the score distributions of target and non-target speakers overlap only minimally.

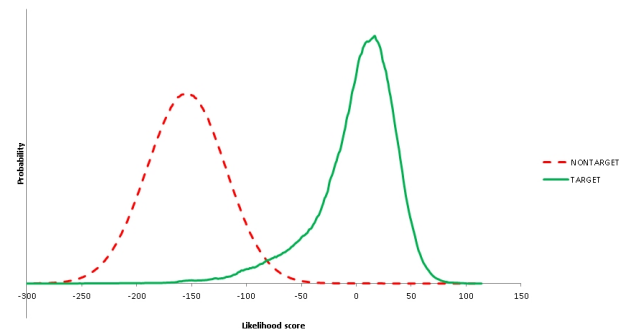


Fig. 1. Score distributions for target and non-target speakers with the VoxForge test set.

The reliability of speaker verification on a given test data is generally presented on the Detection Error Trade-off (DET) Curve [14]. DET-curve plotting software provided by NIST [15] was used to create the graph of the DET curve of the speaker verification task. Equal Error Rate (EER) is about 3%.

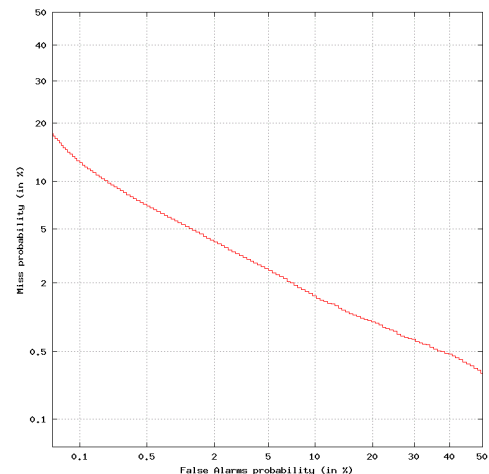


Fig. 2. DET curve of the speaker verification tested on VoxForge speech database

Preliminary tests of the Authorization task were done on a very small database of 10 speakers and 410 test samples, recorded in the environment of the DLR's Air Traffic Control

Simulator. The results presented on Fig. 3. show how the error is increased with the number of authorized persons.

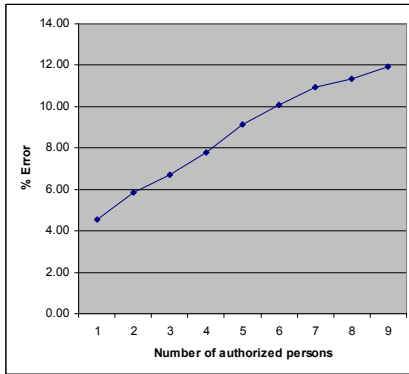


Fig. 3. Results of the preliminary evaluation of the influence of the number of authorized speakers on the reliability of authorization.

B. Stress Detection

A number of research works examine the presence of emotions or stress in speech [16][17][18]. Others study specific characteristics of speech under stress [19][20]. In all these articles the choice of research material, methodology, and even definitions were determined by the intended use of the results and the domain of their future application. This work is focused on the stress, which is reflected into changes of arousal and therefore also to the measurable speech characteristics.

In general, stress can be understood as a real or implied threat to the psychological or physiological integrity of an individual [21]. The notion of stress covers a very wide range of phenomena and their effects on the speech signal are highly non-specific (among others also Lombard speech [22]). Based on the psychological understanding of concepts of stress, emotion and arousal, we decided to operationalize stress induced changes in speech as changes in arousal.

Speaker independent on-line stress/arousal level detector from speech was designed using KALDI [9]; Gaussian Mixture Modeling was used [23], and models of speech with different levels of arousal were created by adapting the universal model. The adaptation database of acted speech with various levels of arousal was recorded following the methodology used in [24] (15 speakers, 7 minutes of speech per arousal level).

Three levels of arousal - neutral, increased, and high (levels 1, 2, 3) were considered in the first version. The classification accuracy is expressed in terms of recall and precision (see TABLE 1). Recall is defined as the number of true positive decisions (stress level correctly identified) over the number of true positives plus the number of false negatives (stress incorrectly rejected). Precision is defined as the number of true positive decisions over the number of all positive decisions.

TABLE 1. Test results of identification of three stress levels on 553 utterances for each level from 7 different speakers

	Stress Level 1	Stress Level 2	Stress Level 3
Recall	92.77%	51.18%	75.59%
Precision	72.97%	64.46%	80.85%

For more details on the stress detector see [25].

IV. CORRELATION WITH NON-VOCAL MODALITIES

These two vocal modalities are now correlated with two additional non-vocal modalities, i.e. aircraft conformance monitoring as well as aircraft conflict detection algorithms. Both functions require the availability of reliable surveillance data as well as reliable information about valid ATC clearances given by the controller.

Sources for surveillance data can be primary and secondary radar systems (PSR / SSR), multilateration systems (MLAT) or surveillance systems using Automatic Dependent Surveillance – Broadcast (ADS-B) [26].

There are also several sources for ATC clearance data available, such as:

- Electronic Flight Strips (EFS): Flight strips are used to document issued clearances and help to maintain the actual traffic picture. Electronic Flight Strips are already state of the art [27], [28], [29].
- Mouse input: Some Air Navigation Service Providers use radar systems, which allow the input of ATC clearances into the radar label [30].
- Speech recognition: Recent research activities used speech recognition to obtain the given ATC clearance directly from the controller's utterances [31].

A. Conformance monitoring

Conformance monitoring is in principle the continuous comparison of the aircraft manoeuvres with its expected behavior, which is defined by the given ATC clearances. The purpose is to immediately detect deviations from issued ATC clearances and to generate situational awareness of the air traffic controller. In case of successfully infiltrated fake ATC clearances, this function would give an alert as soon as the pilot executes this false advice, because this fake ATC clearance is not captured as a valid clearance by the information sources listed above.

Conformance monitoring tools are already about to become a standard CWP functionality; system and functional requirements are already regulated [32]. Nevertheless conformance monitoring has not yet been used in the frame of ATM security management.

The variety of possible ATC clearances strongly depend on local attributes such as the airspace structure, local approach and departure procedures and local operating procedures. The most important feature is the category of the ATC unit which uses conformance monitoring functions (Tower, Approach, Area control), because the corresponding ATC procedures are very different [2].

Within the GAMMA project it was decided to start with Approach and Area Control procedures, which leads to the following definition of classes of conformance:

- Level conformance,

- Lateral conformance,
- Horizontal speed conformance,
- Vertical speed conformance,
- Trajectory conformance.

Following types of non-conformance have been defined:

- 1) The aircraft does not respond to a given ATC clearance,
- 2) The aircraft responds to a given ATC clearance, but the instructed maneuver is not executed correctly,
- 3) The aircraft initiates any maneuver without being cleared by ATC.

Reaction times have to be considered for conformance monitoring, as the pilot has to confirm the ATC clearance and has to operate the autopilot or the flight controls. Additionally, a certain time period may be necessary to complete the action instructed with an ATC clearance. Several defined tolerances have also to be considered, e.g. an allowed vertical deviation from the allocated level by 200ft [2] or performance standards for navigation [33]. The figure below gives an example for level conformance monitoring; corresponding examples for all three types of non-conformance are displayed.

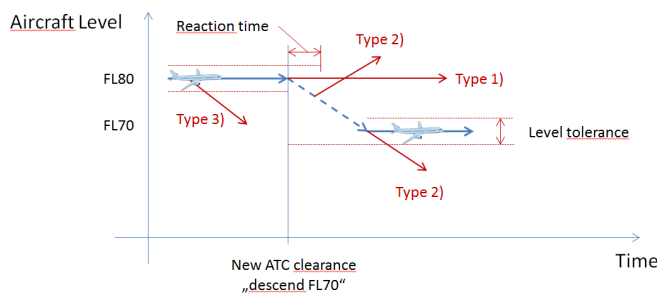


Fig. 4. Level conformance monitoring and types of non-conformance

B. Conflict detection

A conflict detection function continuously monitors all aircraft in a defined area and predicts conflicts between two or more aircraft. For IFR traffic, a conflict is given when the applicable minimum vertical or lateral separation is not ensured. Therefore a risk of a collision exists [2].

Conflict detection functions are already a common standard for ATC surveillance systems. EUROCONTROL distinguishes between medium-term conflict detection (MTCD) [34] and short-term conflict alert (STCA) [35]. MTCD uses planned trajectories and indicates conflicts up to 20 minutes in advance while STCA mainly uses specific aircraft state vectors with a pre-notification time of up to 2 minutes.

From the controller's point of view, MTCD can be understood as a tool supporting and rechecking the current traffic preplanning while STCA is an additional safety net to immediately prevent collisions. Although it is primarily a safety issue, a detected conflict between two aircraft can be one indicator for successfully infiltrated unauthorized fake ATC clearances.

C. Overall threat indicator

Individual indicators are additionally correlated to an overall threat indicator score as a basis for automatic reporting. These reports can be displayed to the user (situational awareness) or can be forwarded to security management instances, responsible authorities, units responsible for air traffic flow and capacity management (ATFCM) and other ATM stakeholders.

As a short summary, the following indicators are obtained from the single modalities:

- For each analyzed voice communication transmission: Speaker ID (if known), speaker verification score, stress score, timestamp
- For each detected non-conformance, as long as the non-conformance is valid: Aircraft ID, class of non-conformance, valid from, valid to
- For each detected conflict, as long as the conflict exists: Aircraft IDs, closest predicted proximity, time until closest predicted proximity, valid from, valid to.

The following procedure is used to correlate these indicators and to obtain the overall threat indicator score:

- 1) Add timestamps between 'valid from' and 'valid to' following a defined time interval
- 2) Define a time frame that shall be used for correlation (e.g. from now until x minutes in the past)
- 3) Define minimum thresholds for speaker verification and stress detection scores, which must be reached to be considered as relevant
- 4) Count all indications within the defined time frame separately for every modality (intermediate scores)
- 5) Use weighting factors on these intermediate scores separately for every modality. Weighting factors should consider the individual module reliability and must be further adjusted to the ATC environment (typical amount of air traffic, typical controller workload, use of data link etc.)
- 6) Sum up to an overall threat indicator score
- 7) Define a threshold that must be reached to trigger an alert and automatic reporting.

The expected behavior of the overall threat indicator score during an ongoing attack is sketched in the following figure.

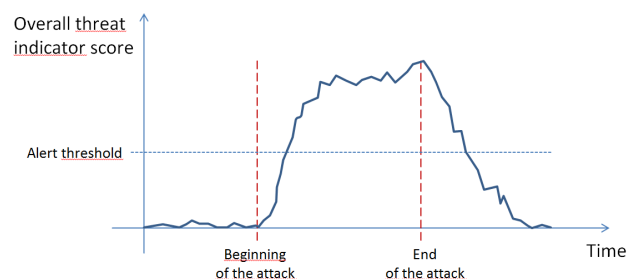


Fig. 5. Expected behaviour of the overall threat indicator during a security incident in ATC voice communication

V. VALIDATION METHODOLOGY

The validation campaign of the SACom prototype takes place from April to October 2016. It involves several iterations of a defined set of real-time human in the loop simulations by making use of DLR's Air Traffic Management and Operations Simulator (ATMOS) located in Braunschweig and an open-source VoIP radio communication simulator called Yada. Trained pseudo pilots will communicate with the test person in the role of the air traffic controller and will steer the simulated aircraft.

The simulated ATC environment is Düsseldorf Approach; a standardized realistic traffic scenario is used. The SACom prototype is installed at the controller working position.

A single iteration includes the following steps:

- 1) Introduction / Briefing
- 2) The controller is added to the database of authorized speakers by creating a voice enrollment
- 3) ATMOS Training simulation. This is to get used to the simulator. Also the created voice enrollment is verified.
- 4) 20 short-term simulations (3-6 minutes per simulation) with predefined non-conformance / conflict situations; the controller should tactically react to these events; the SACom system is running in the background but no indications are displayed ('baseline' situation)
- 5) SACom Training simulation: This is to get used with the SACom indications. Due to a very low workload it is expected that the stress level of the controller will be very low. This simulation is therefore used to measure the 'baseline' situation for the stress detection function.
- 6) Attack scenario: this simulation is planned for a duration of 40 minutes of an average workload. During the first 10 minutes no events will be simulated. Afterwards an unauthorized person will intrude the ATC voice communication and will insert fake ATC clearances with the goal to cause critical conflicts or at least disturb the flow of traffic (about 1 fake clearance every 2 minutes). After 30 minutes of simulation the attack will cease. The controller will in general have the task to use the SACom system to tactically react to the intrusion and to maintain flight safety and traffic capacity as long as possible.
- 7) Feedback and questionnaires regarding acceptance, situational awareness, trust and human factors.

The result analysis will include specific measurements to defined key performance areas (see [36]) as well as qualitative assessments. The overall validation procedure is based on the European Operational Concept Validation Methodology (E-OCVM) [37].

VI. FIRST RESULTS AND DISCUSSION

Until now the validation activities showed, that the speaker verification can in principle be used for typical ATC voice communication. Under laboratory conditions with a good audio

quality, fair to good recognition rates were achieved. It was discovered that the correct splitting of controller's and pilot's utterances for separate analysis can be a challenge; this problem is not yet completely solved. Another critical factor is the speed of the voice analysis. The first versions of the speaker verification module had a delay of several seconds until the results were displayed. This makes it more difficult for the controller to allocate the displayed result to a transmission of himself or another transmission which is already some seconds in the past.

For the stress detection function it was very difficult to induce realistic stress reactions in the simulation on one hand and to measure a difference compared to the baseline on the other hand. Controllers are obviously used to stressful situations and they are trained not to show stress. Also the pseudo pilots showed no stress at all due to their training status and due to the fact that they have to be familiar with the whole simulated scenario.

The conformance monitoring has been found to have the potential to detect attack effects much faster than a human operator, depending on the actual workload. The surprise effect and controller's expectations play a very important role.

Regarding the conflict detection, the amount of data collected until the publication of this paper is not yet representative to discuss it. As it is the controller's profession to prevent collisions between two aircraft, there were only very few conflicts actually happening during overload situations. In all other cases the reaction of the controller was prompt to solve the problem in time.

Two of the used four modalities did, until the publication of this paper, not yet produce a significant effect on the overall threat indicator; gathered data are not yet representative to discuss them and to draw any conclusions.

Until the publication of this paper, the validation activities showed that the speaker verification as well as the conformance monitoring can directly detect the intrusion into ATC voice communication by an unauthorized speaker or the effects of fake ATC clearances.

Due to their nature or their actual reliability, the stress detection and the conflict detection functions are less suitable to be used directly as detection functions, but may nevertheless be used as additional parameters (stress detection) or as backup safety net (conflict detection). For example the speaker verification scores showed in general lower values in high stress situations, which may allow the reduction of false alarms due a stressed controller's speech, provided that the stress detection is sensitive enough to detect a stress reaction that is typical for ATC voice communication.

The weak point of the described setup is, apart from the speaker verification, that all other functionalities are in principle able to detect, but are not able to identify the attack, as also ATM safety issues show exactly the same effects, e.g.:

- A stressed voice may also be caused by an emergency, a training situation, unusual situations etc.,

- Aircraft deviations from the cleared flight path may also be caused by technical failures or pilot errors,
- Conflicts may also happen for other reasons, e.g. due to ATC errors.

As a conclusion, to be able to clearly distinguish an attack from safety events, more independent indicators beside the described ones have to be used; their reliability has to be maximized.

Nevertheless, at the moment it is still unavoidable that a human operator analyses the situation and verifies the indication as soon as an alert is raised. Also the tactical treatment still rests with the operator. Therefore, beside the equipment, an appropriate training is essential.

As the development of the SACom system and the related validation activities will continue, it can be expected that complete results and conclusions will be published in the year 2017.

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More information can be found under www.gamma-project.eu.

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